Unit C6L

SKU:U202









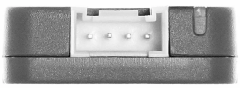
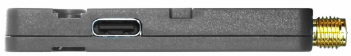










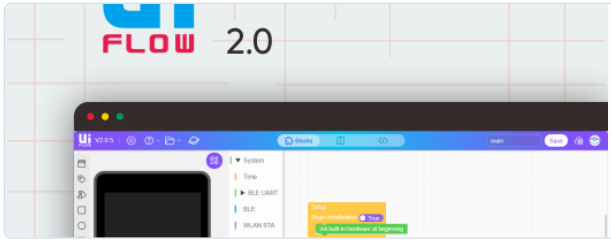







Description

Unit C6L is a highly integrated **LoRa** communication unit, equipped with the **Stamp C6LoRa** module (featuring **ESP32-C6** as the core controller, integrating **SX1262** LoRa transceiver and RF switch circuit), supporting **868 ~ 923 MHz** operating frequency range. Unit C6L integrates a **0.66" OLED** monochrome display, programmable RGB indicator, buzzer, and user button, providing intuitive human-machine interaction. The device supports connection to the **Meshtastic** network or secondary development, enabling rapid deployment of private LoRa networks, suitable for remote monitoring, outdoor exploration, and emergency communication scenarios. In hardware design, Unit C6L offers 2x RP-SMA antenna interfaces, respectively for **2.4 GHz Wi-Fi 6** and **LoRa** communication, ensuring scalability and stability of wireless performance. It also integrates **over-voltage protection chip AW32901FCR** to enhance device stability. With HY2.0-4P expansion interface and LEGO-compatible mounting holes, users can easily expand sensors or modules to quickly build more application scenarios.

Tutorial



UiFlow2

This tutorial introduces how to control Unit C6L through the UiFlow2 graphical programming platform.



Arduino IDE

This tutorial explains how to program and control Unit C6L using Arduino IDE.



Meshtastic Tutorial

This tutorial introduces how to use Meshtastic with Unit C6L.

Features

- Supports Meshtastic
- ESP32-C6 core controller
- 2.4 GHz Wi-Fi 6
- SX1262 LoRa transceiver
- LoRa operating frequency: 868 ~ 923 MHz
- Programmable RGB LED WS2812C
- SSD1306 0.66" OLED monochrome display, resolution 64x48
- Human-machine interaction:
  - Buzzer
  - User button
- Integrated over-voltage protection chip AW32901FCR
- HY2.0-4P expansion interface
- LEGO-compatible mounting holes

Includes

- 1 x Unit C6L
- 1 x 2.4 GHz RP-SMA whip antenna
- 1 x 868 MHz RP-SMA whip antenna
- 1 x HY2.0-4P Grove Cable (20cm)

Applications

- Smart agriculture
- Industrial monitoring
- Outdoor communication

Specifications

Specification	Parameter
SoC	ESP32-C6 (RISC-V 32-bit high-performance single-core processor 160MHz + RISC-V 32-bit low-power single-core processor 20 MHz)
Flash	16MB
Display	SSD1306 0.66" OLED monochrome screen, resolution 64 x 48
LoRa	SX1262
Antenna Specs	Wi-Fi Antenna: length 84mm, RP-SMA connector (female thread, female pin), operating frequency 2.4 GHz, gain 3dBi LoRa Antenna: length 108mm, RP-SMA connector (female thread, female pin), operating frequency 868 MHz, gain 3dBi
Max LoRa TX Power	+22dBm
Power Consumption	Sleep mode (HY2.0-4P powered): 696.86uA Sleep mode (USB Type-C powered): 866.42uA LoRa RX standby (USB Type-C powered): 85.18mA LoRa max power continuous TX (USB Type-C powered): 80.02mA
LoRa RX Sensitivity	-147 dBm (LoRa low data rate mode)
Power Input	DC 5V (USB Type-C)
RGB LED	1x WS2812C
Product Size	62.0 x 24.0 x 8.0mm (excluding antenna)
Product Weight	13.6g (excluding antenna)
Package Size	104.0 x 44.0 x 12.5mm
Gross Weight	38.6g

Learn

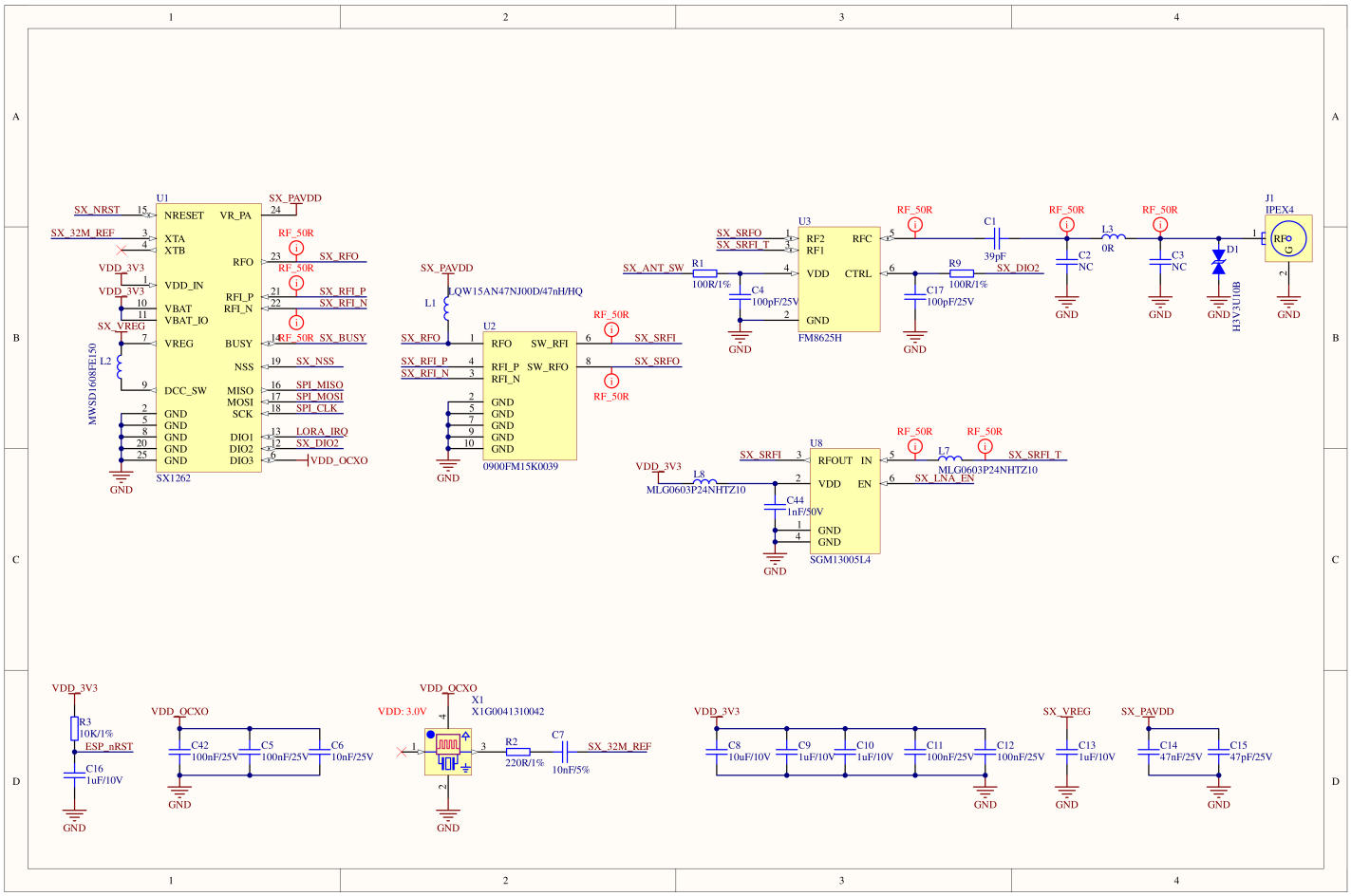
Download Mode

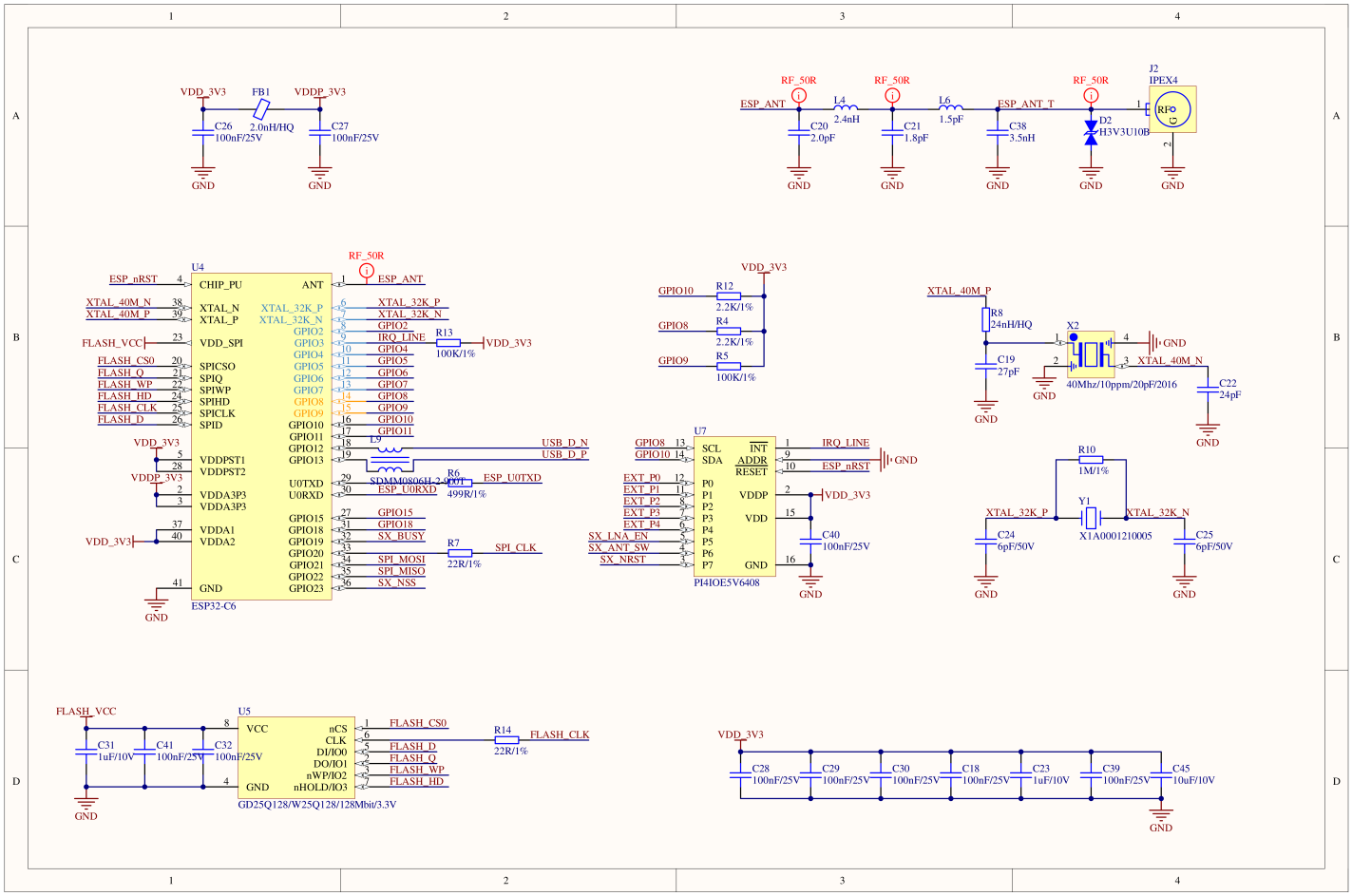
To flash firmware, connect the device to a computer via a USB Type-C data cable, press and hold the side reset button for 3 seconds until the green light turns red. At this point, the device enters download mode and is ready for flashing.

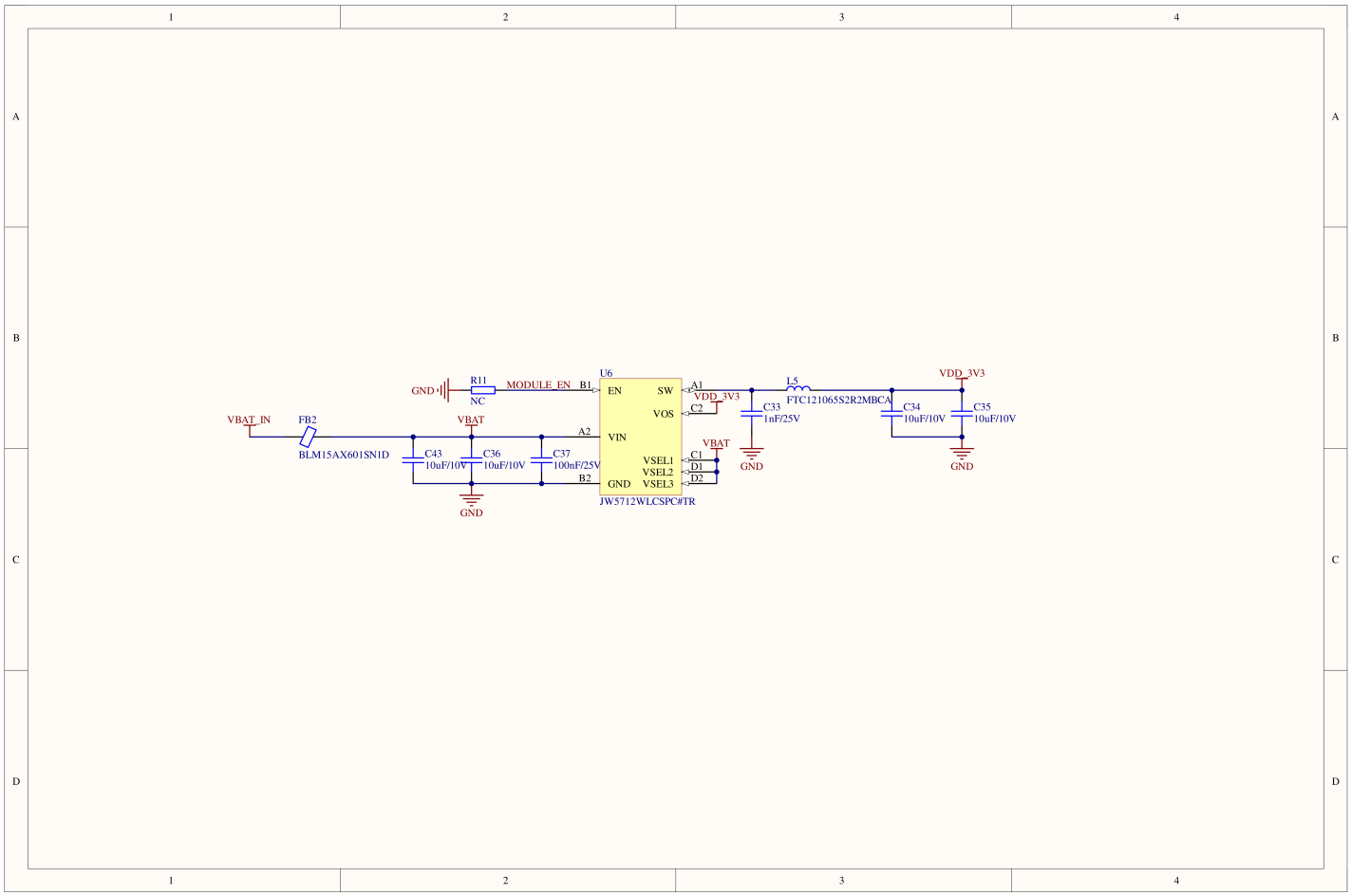


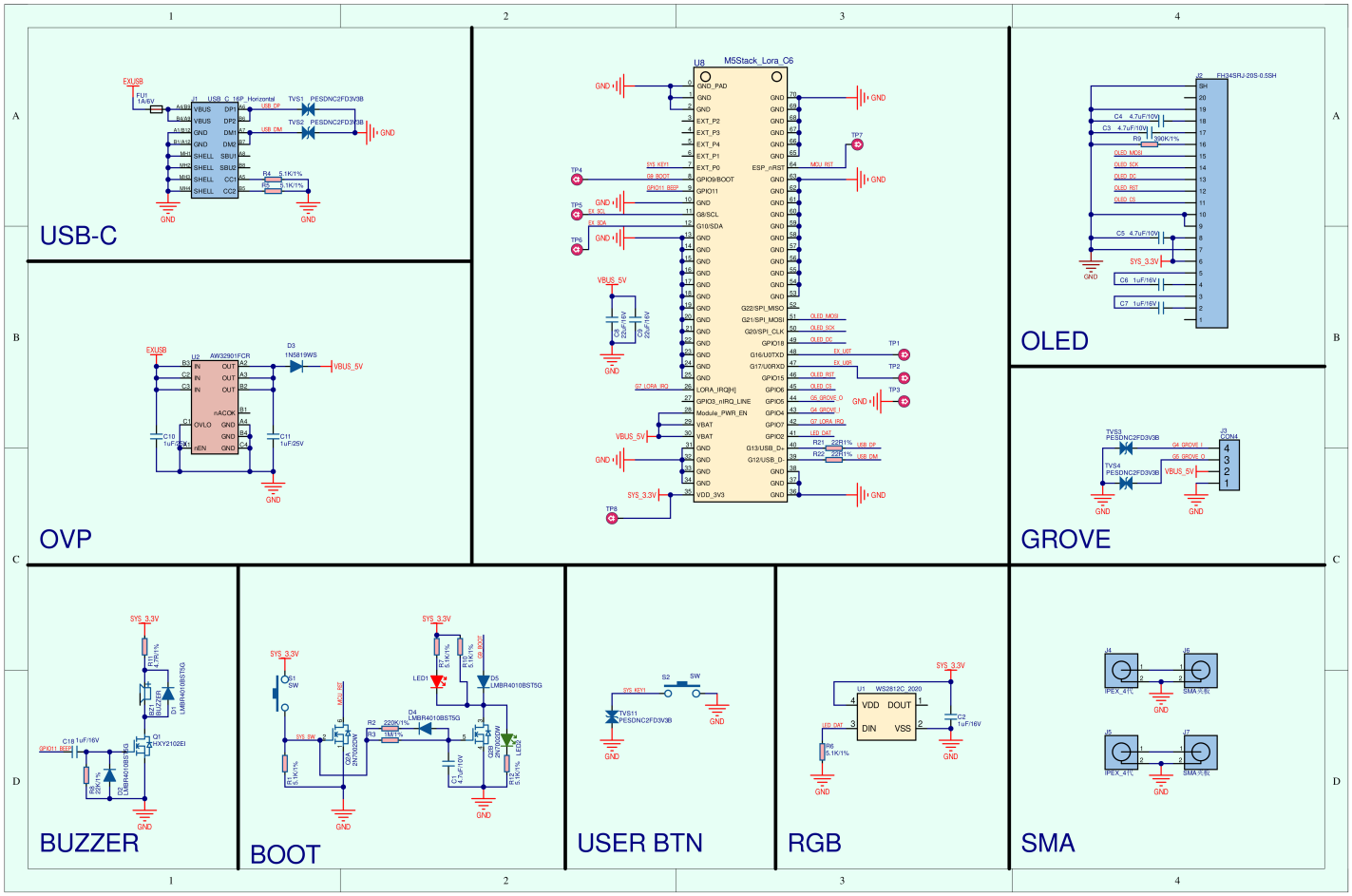
Schematics

- Unit C6L Schematics PDF
- Stamp C6LoRa Schematics PDF




VDD_3V3
FB1
VDDP_3V3
C26
100nF/25V
2.0nH/HQ
C27
100nF/25V
GND
RF_50R
ESP_ANT
L4
2.4nH
L6
1.5pF
ESP_ANT_T
C20
2.0pF
C21
1.8pF
C38
3.5nH
D2
H3V3U10B
J2
IPEX4
RF
U4
CHIP_PU
ANT
XTAL_N
XTAL_P
VDD_SPI
SPICSO
SPIQ
SPIWP
SPIHD
SPICLK
SPID
VDDPST1
VDDPST2
VDDA3P3
VDDA3P3
VDDA1
VDDA2
GND
ESP32-C6
ESP_nRST
XTAL_40M_N
XTAL_40M_P
FLASH_VCC
FLASH_CS0
FLASH_Q
FLASH_WP
FLASH_HD
FLASH_CLK
FLASH_D
XTAL_32K_P
XTAL_32K_N
GPIO2
IRQ_LINE
R13
100K/1%
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
L9
USB_D_N
USB_D_P
R6
SDMM0806H-2.4NT
U0TXD
U0RXD
ESP_U0TXD
ESP_U0RXD
499R/1%
GPIO15
GPIO18
SX_BUSY
R7
22R/1%
SPI_CLK
GPIO20
GPIO21
SPI_MOSI
SPI_MISO
GPIO22
GPIO23
SX_NSS
R12
2.2K/1%
R4
2.2K/1%
R5
100K/1%
GPIO10
GPIO8
GPIO9
U7
SCL
SDA
INT
ADDR
RESET
P0
P1
P2
P3
P4
P5
P6
P7
VDDP
VDD
PI4IOE5V6408
EXT_P0
EXT_P1
EXT_P2
EXT_P3
EXT_P4
SX_LNA_EN
SX_ANT_SW
SX_NRST
IRQ_LINE
ESP_nRST
C40
100nF/25V
R8
24nH/HQ
C19
27pF
X2
40Mhz/10ppm/20pF/2016
C22
24pF
R10
1M/1%
Y1
X1A0001210005
C24
6pF/50V
C25
6pF/50V
U5
VCC
CLK
DI/IO0
DO/IO1
nWP/IO2
GND
nCS
nHOLD/IO3
GD25Q128/W25Q128/128Mbit/3.3V
R14
22R/1%
C31
1uF/10V
C41
100nF/25V
C32
100nF/25V
C28
100nF/25V
C29
100nF/25V
C30
100nF/25V
C18
100nF/25V
C23
1uF/10V
C39
100nF/25V
C45
10uF/10V


R11
NC
MODULE_EN
U6
EN
SW
VOS
VIN
VSEL1
VSEL2
VSEL3
GND
JW5712WLCSPC#TR
VBAT_IN
FB2
BLM15AX601SN1D
VBAT
C43
10uF/10V
C36
10uF/10V
C37
100nF/25V
C1
C2
D1
D2
L5
FTC121065S2R2MBCA
C33
1nF/25V
C34
10uF/10V
C35
10uF/10V
VDD_3V3
GND



PinMap

Stamp C6LoRa

ESP32-C6	G21	G22	G20	G23	G19	G7
SX1262	MOSI	MISO	SCK	CS	BUSY	IRQ

ESP32-C6	G8	G10	G3	ESP_RST
PI4IOE5V6408	SCL	SDA	INT	RST

PI4IOE5V6408	E0.P5	E0.P6	E0.P7
SX1262	SX_LNA_EN	SX_ANT_SW	SX_NRST

The following steps are required for LoRa module initialization:

- Reset the LoRa module: Set SX_NRST low, hold for 100 ms, then set high.
- Enable the RF antenna switch: Set SX_ANT_SW high.
- Enable the LNA chip: Set SX_LNA_EN high.

OLED

ESP32-C6	G6	G21	G20	G18	G15
SSD1306	CS	MOSI	SCK	DC	RST

HMI

ESP32-C6	G11	G2
Buzzer	BEEP	
RGB LED		LED_DAT

PI4IOE5V6408	P0
SYS_KEY1	INPUT

USB

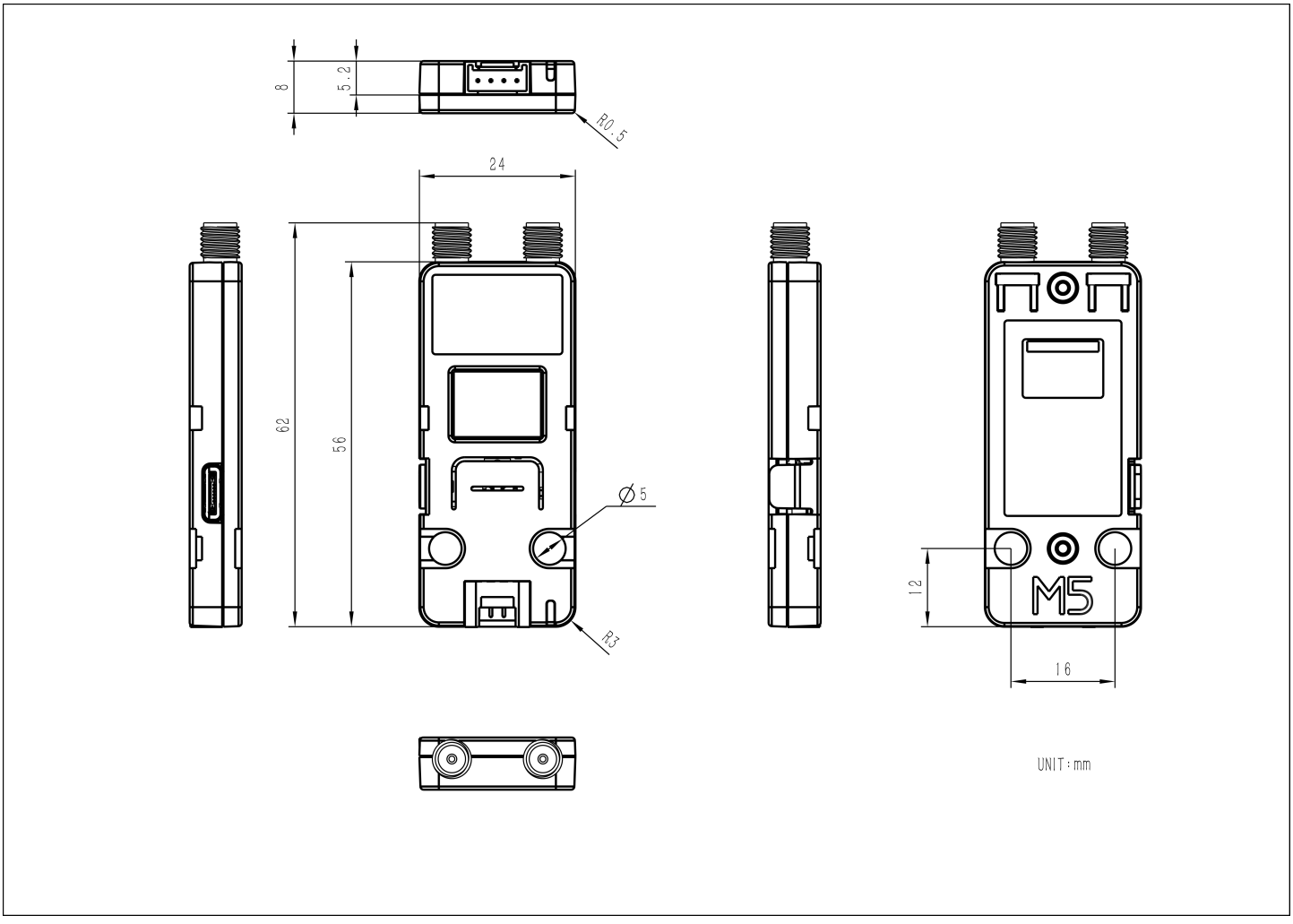
ESP32-C6	G13	G12
USB	USB_D+	USB_D-

HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	G5	G4

Model Size

- Unit C6L Model Size PDF



Structure

- Unit C6L Structure Files

Datasheets

- SX1262
- ESP32-C6

Softwares

Quick Start

- Use Meshtastic via Unit C6L

Arduino

- Unit C6L Arduino Quick Start

UiFlow2

- Unit C6L UiFlow2 Quick Start

PlatformIO

```
[env:m5stack-unitc6l]
platform = https://github.com/pioarduino/platform-espressif32/archive/refs/heads/develop.zip
board = esp32-c6-devkitc-1
framework = arduino
upload_speed = 1500000
monitor_speed = 115200
build_flags =
    -D ARDUINO_USB_MODE=1
    -DARDUINO_USB_CDC_ON_BOOT=1
    -DCONFIG_ARDUHAL_LOG_COLORS=1
    -DCORE_DEBUG_LEVEL=5
lib_deps =
    M5Unified=https://github.com/m5stack/M5Unified
    RadioLib = https://github.com/jgromes/RadioLib
```

EasyLoader

Easyloader	Download	Note
Unit C6L User Demo Easyloader	download	/

Note

Please notice that this firmware isn't the newest version, mqtt and webserver was disabled by default. This firmware is for testing whether the Unit C6L is working correctly. If you need to use Meshtastic, please go to the official website to flash the latest version of the firmware: https://flasher.meshtastic.org/

Video

- Unit C6L Product Introduction and Function Demonstration

U202_Unit-C6L-video.mp4